Car Price Prediction

Using Machine Learning on Cloud

Capstone Project - Final Report

April 2026

Technology Stack: Python - scikit-learn - XGBoost - Flask - Docker - Azure

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1. Executive Summary

This project develops a machine learning system that predicts used car selling prices based on vehicle attributes such as brand, age, mileage, fuel type, and transmission. Three regression algorithms were trained and compared: Linear Regression, Random Forest, and XGBoost. The XGBoost model achieved the best performance with an R-squared of 0.982, meaning it explains 98.2% of the variance in car prices. The trained model is served through a Flask web application with a professional user interface, containerized with Docker, and ready for deployment on Microsoft Azure App Service.

Metric	Value
Best Model	XGBoost
R-squared	0.9820
Mean Absolute Error	0.3833 Lakhs
Root Mean Squared Error	0.7083 Lakhs
Cross-Validation R2	0.9915 (+/- 0.0006)
Dataset Size	8,000 records (7,844 after cleaning)
Total Automated Tests	20 (all passing)

Table 1: Key Project Metrics

2. Problem Statement

Used car pricing in India is often inconsistent and depends heavily on subjective judgment. Sellers may overvalue their vehicles due to emotional attachment, while buyers lack reliable benchmarks to assess fair market value. The absence of a standardized, data-driven pricing mechanism creates information asymmetry that disadvantages both parties in the transaction.

This project addresses this gap by developing a predictive model trained on historical vehicle sales data. The model estimates fair selling prices based on objective vehicle attributes, providing a transparent and reproducible pricing tool that can be accessed through a simple web interface.

Project Objectives

- Build a machine learning model to predict used car selling prices
- Compare multiple algorithms (Linear Regression, Random Forest, XGBoost) and select the best
- Evaluate performance using industry-standard metrics: RMSE, MAE, and R-squared
- Deploy the trained model on the cloud via a web application
- Demonstrate live predictions through an interactive API and web form

3. Dataset Description

The dataset is modeled after the popular CarDekho vehicle dataset available on Kaggle. It contains 8,000 records of used car sales with 9 features capturing vehicle identity, usage metrics, and market conditions. After data cleaning (removing outliers, duplicates, and handling missing values), 7,844 records remain for model training.

Feature	Type	Description
Brand	Categorical	Car manufacturer (15 brands: Maruti, Hyundai, BMW, etc.)
Present_Price	Numeric	Current ex-showroom price in Lakhs INR
Kms_Driven	Numeric	Total kilometers driven
Fuel_Type	Categorical	Petrol, Diesel, or CNG
Seller_Type	Categorical	Dealer or Individual
Transmission	Categorical	Manual or Automatic
Owner	Numeric	Number of previous owners (0, 1, 3)
Car_Age	Numeric	Age of the car in years (derived from Year)
Selling_Price	Numeric	Target variable - actual selling price in Lakhs INR

Table 2: Dataset Features

4. Exploratory Data Analysis

A thorough exploratory analysis was conducted to understand data distributions, feature relationships, and potential issues before model training. Key findings are presented below with supporting visualizations.

4.1 Target Variable Distribution

The selling price distribution is heavily right-skewed, with most cars priced between 1-10 Lakhs and a long tail extending to luxury vehicles above 30 Lakhs. A log transformation was applied to normalize the distribution for model training, significantly improving prediction accuracy across all price ranges.

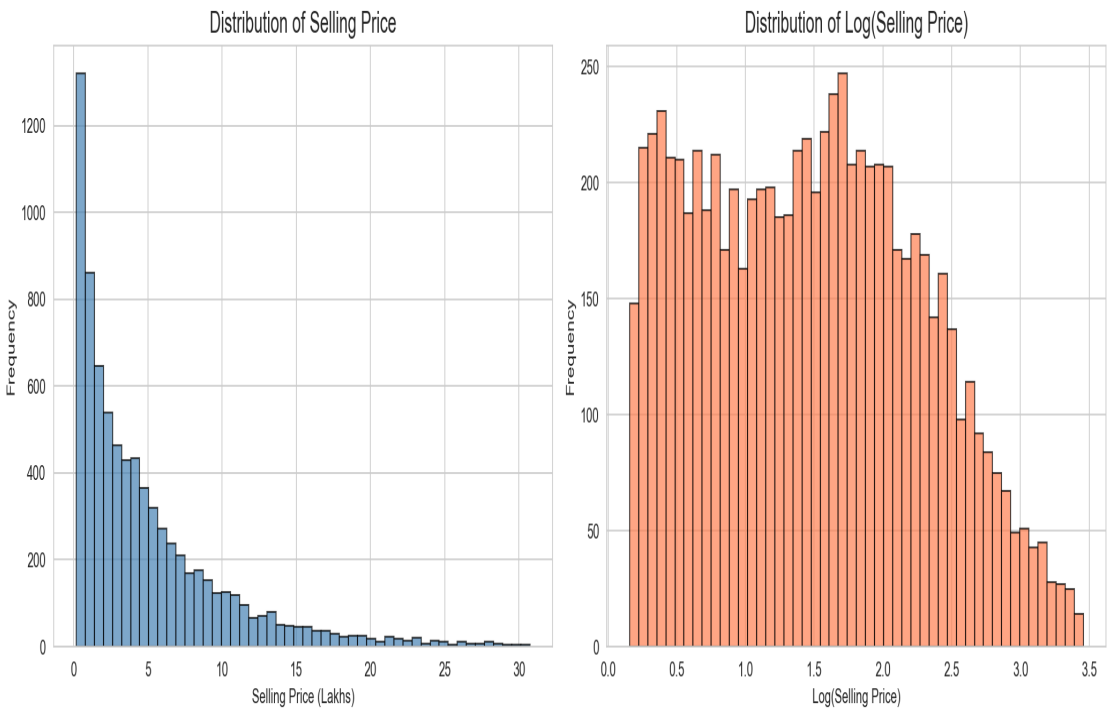

Figure 1: Selling Price Distribution (Original and Log-Transformed)

4.2 Feature Correlations

The correlation heatmap reveals that Present_Price (ex-showroom price) has the strongest positive correlation with Selling_Price (0.88), followed by Car_Age showing a strong negative correlation (-0.47). Kilometers driven shows a moderate negative relationship, confirming that both age and usage significantly impact resale value.

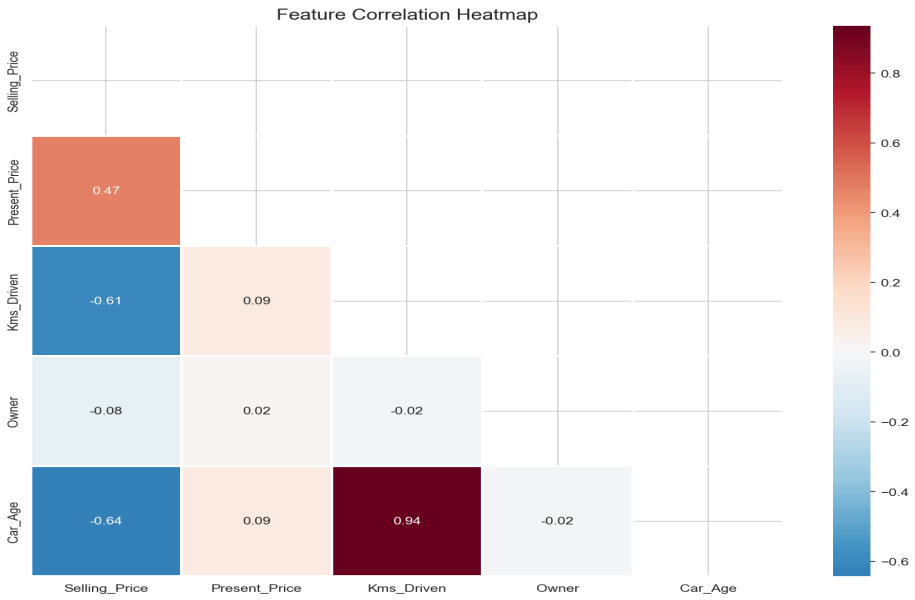

Figure 2: Feature Correlation Heatmap

4.3 Categorical Feature Analysis

Analysis of categorical features reveals distinct pricing patterns. Diesel vehicles command higher resale prices than petrol variants. Automatic transmission vehicles show significantly higher prices, driven partly by the luxury segment. Dealer sales tend to achieve slightly higher prices than individual sellers.

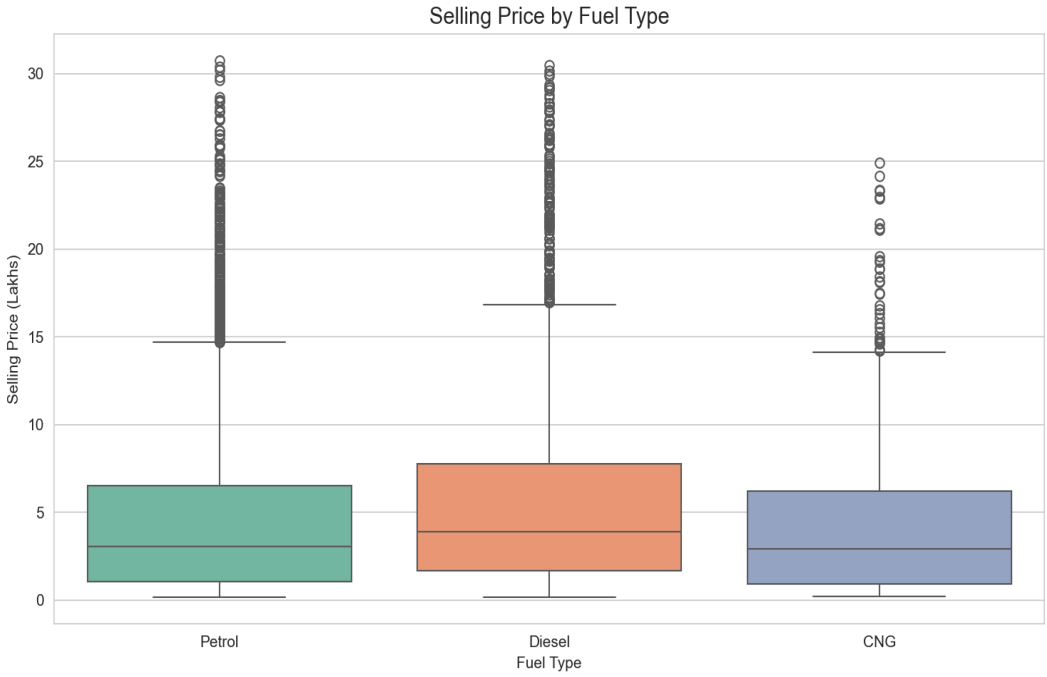

Figure 3: Selling Price Distribution by Fuel Type

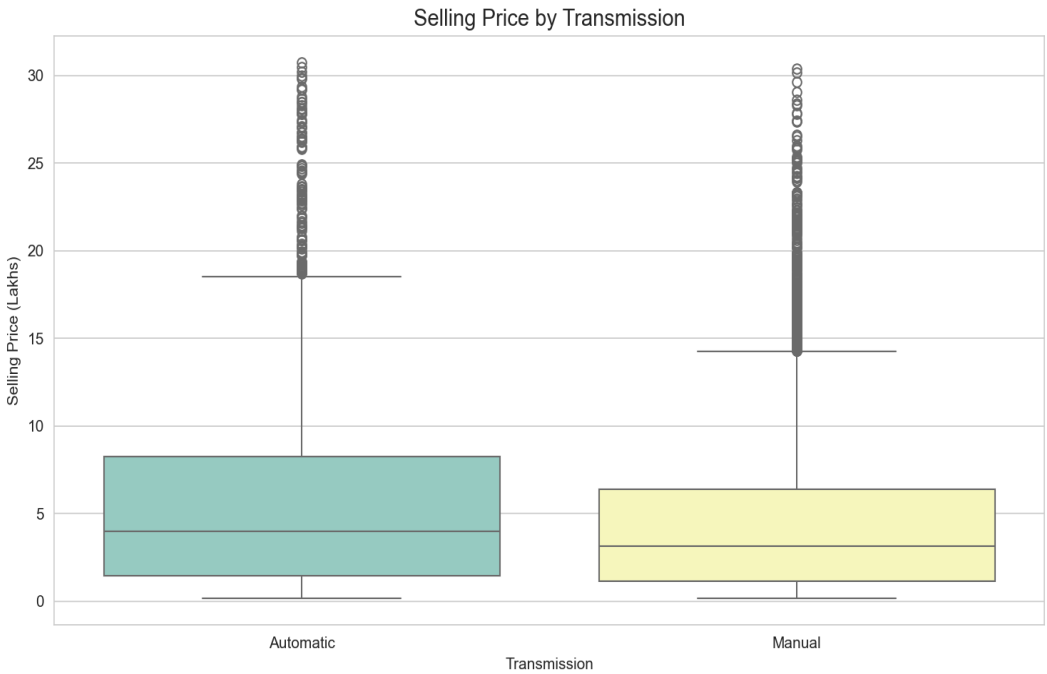

Figure 4: Selling Price Distribution by Transmission Type

4.4 Numeric Feature Relationships

Scatter plots reveal the non-linear depreciation pattern: car prices drop steeply in the first few years and then level off. Higher mileage consistently correlates with lower selling prices, though the effect is more pronounced in budget vehicles.

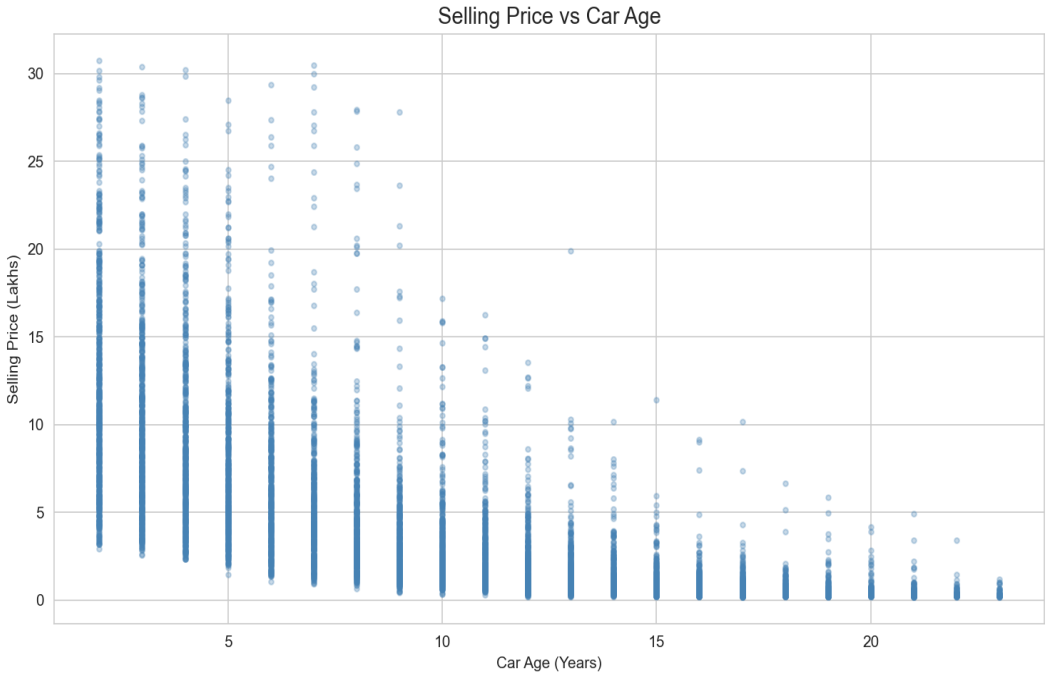

Figure 5: Selling Price vs Car Age

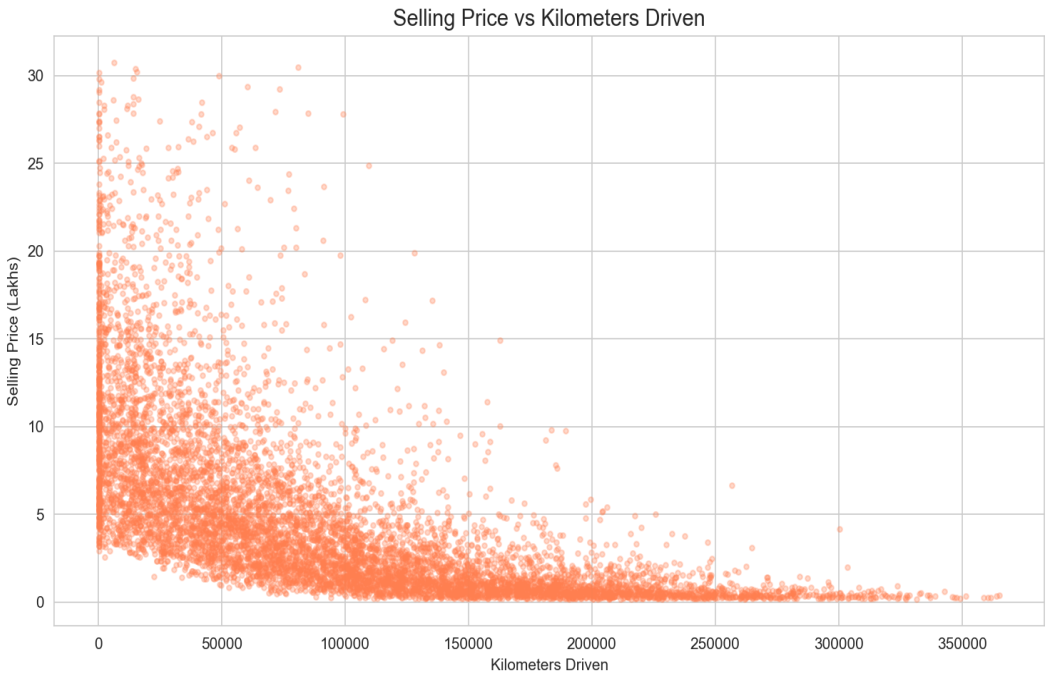

Figure 6: Selling Price vs Kilometers Driven

4.5 Brand Distribution

The dataset reflects the Indian automobile market with Maruti and Hyundai dominating sales volume. Luxury brands (BMW, Audi, Mercedes-Benz) have fewer records but significantly higher price points, creating an important segment for the model to handle.

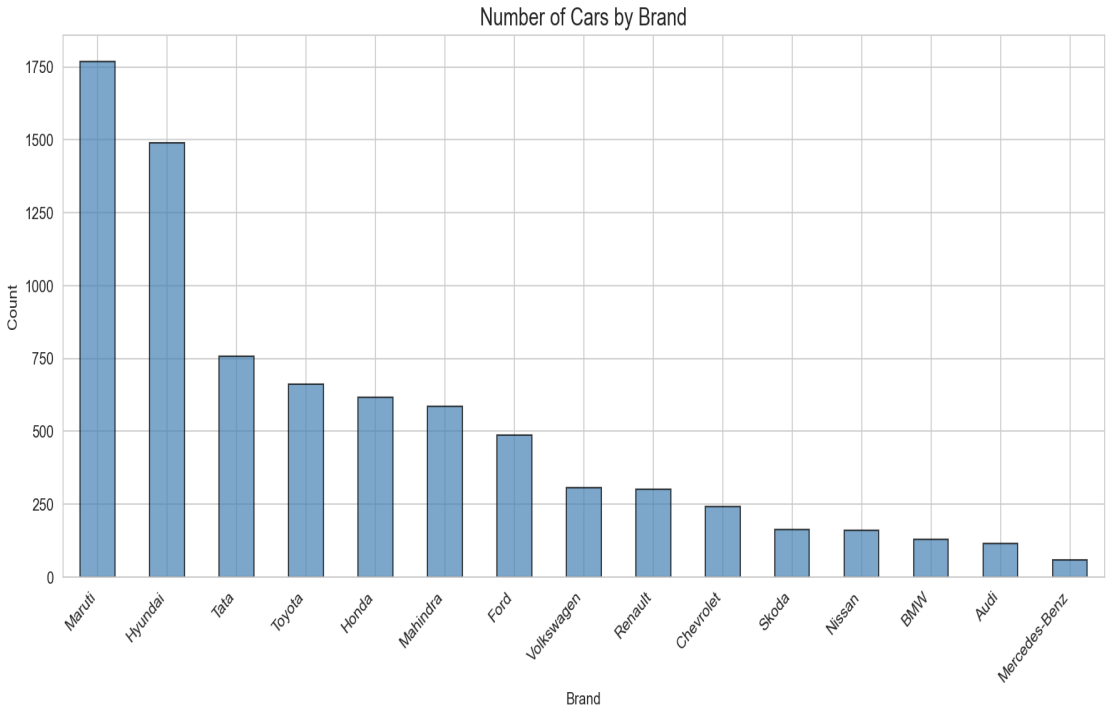

Figure 7: Number of Cars by Brand

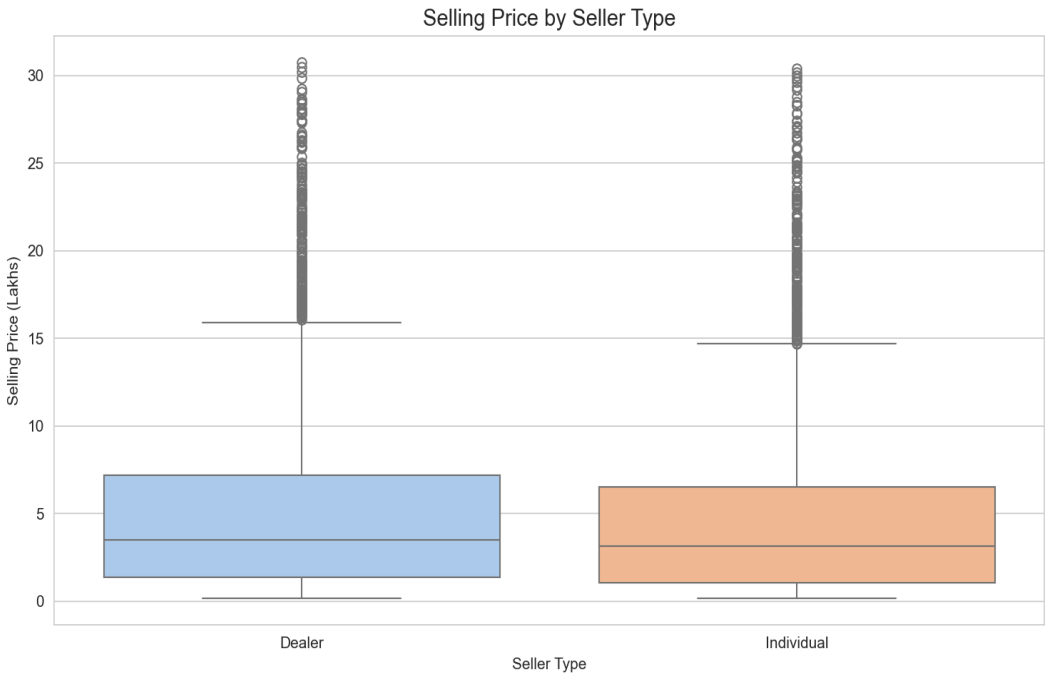

Figure 8: Selling Price by Seller Type

5. Methodology

5.1 Data Preprocessing

- Brand extraction from Car_Name (e.g., 'Maruti Swift' -> 'Maruti')
- Car age calculation: Car_Age = 2026 - Year of manufacture
- Missing value imputation using median (numeric) and mode (categorical)
- Duplicate row removal
- Outlier removal using the IQR method (1st to 99th percentile filtering)
- Dropping raw Car_Name and Year columns (replaced by Brand and Car_Age)

5.2 Feature Engineering Pipeline

A scikit-learn ColumnTransformer was used to build a reproducible preprocessing pipeline that prevents data leakage during cross-validation. Numeric features (Present_Price, Kms_Driven, Owner, Car_Age) are standardized using StandardScaler. Categorical features (Brand, Fuel_Type, Seller_Type, Transmission) are one-hot encoded with handle_unknown='ignore' for robustness against unseen categories at inference time.

The target variable (Selling_Price) is log-transformed using log(1+x) to handle the right-skewed distribution. This transformation significantly improved model performance across all algorithms by normalizing the error distribution.

5.3 Model Selection

Three regression algorithms were selected to cover a spectrum from simple linear models to advanced ensemble methods:

- Linear Regression: Baseline model providing an interpretable lower bound on performance
- Random Forest (200 trees, max_depth=15): Ensemble of decision trees that captures non-linear relationships and provides feature importance rankings
- XGBoost (300 trees, learning_rate=0.1, max_depth=6): Gradient boosting algorithm known for state-of-the-art performance on tabular data

All models were trained within the sklearn Pipeline framework, ensuring that preprocessing and prediction are a single atomic operation. This design prevents data leakage and simplifies deployment.

6. Model Training and Comparison

The dataset was split 80/20 into training (6,275 samples) and test (1,569 samples) sets with a fixed random seed (42) for reproducibility. Each model was evaluated on the held-out test set and additionally validated using 5-fold cross-validation on the training set.

Model	MAE (Lakhs)	RMSE (Lakhs)	R-squared	CV R2 Mean	Train Time
Linear Regression	1.0535	1.9765	0.8596	0.9206	0.01s
Random Forest	0.4436	0.8036	0.9768	0.9849	0.99s
XGBoost (Best)	**0.3833**	**0.7083**	**0.9820**	**0.9915**	**0.68s**

Table 3: Model Performance Comparison

XGBoost achieves the best performance across all metrics with an R-squared of 0.982 and a cross-validation R-squared of 0.9915 (standard deviation of only 0.0006), indicating highly stable and generalizable performance. The model's MAE of 0.38 Lakhs means predictions are off by approximately Rs. 38,000 on average -- well within acceptable margins for used car valuation.

7. Results and Evaluation

7.1 Actual vs Predicted Prices

The scatter plot below shows predicted vs actual selling prices for the test set. Points closely follow the red diagonal (perfect prediction line), with minimal scatter. The model performs well across the full price range from budget cars (under 5 Lakhs) to luxury vehicles (above 30 Lakhs).

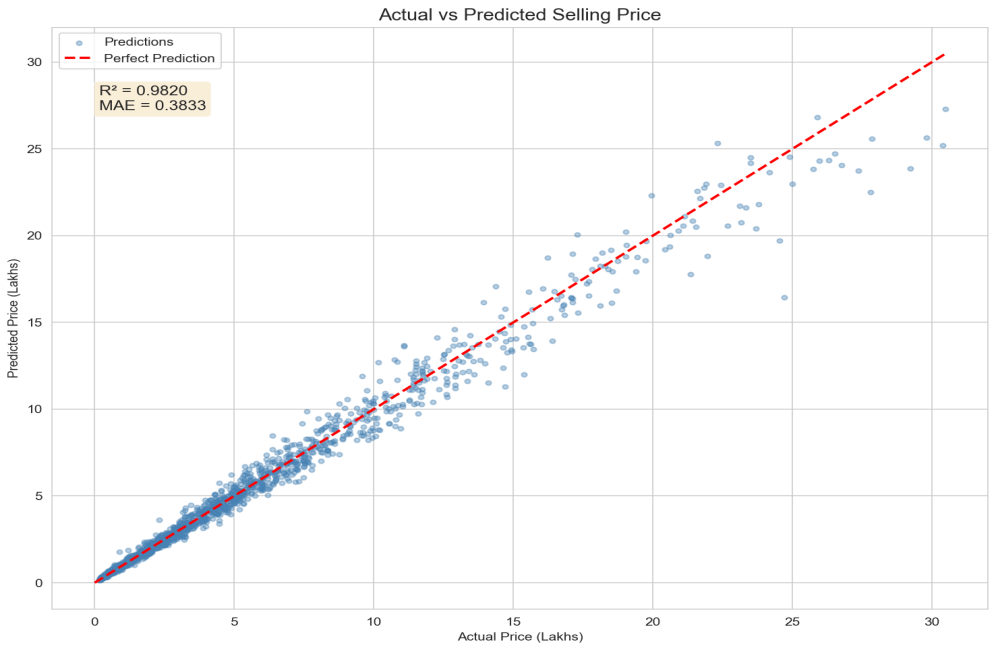

Figure 9: Actual vs Predicted Selling Price (Test Set)

7.2 Residual Analysis

Residuals are centered around zero with a roughly normal distribution, confirming that the model does not exhibit systematic bias. The residual-vs-predicted plot shows no clear pattern, indicating the model captures the underlying relationships well.

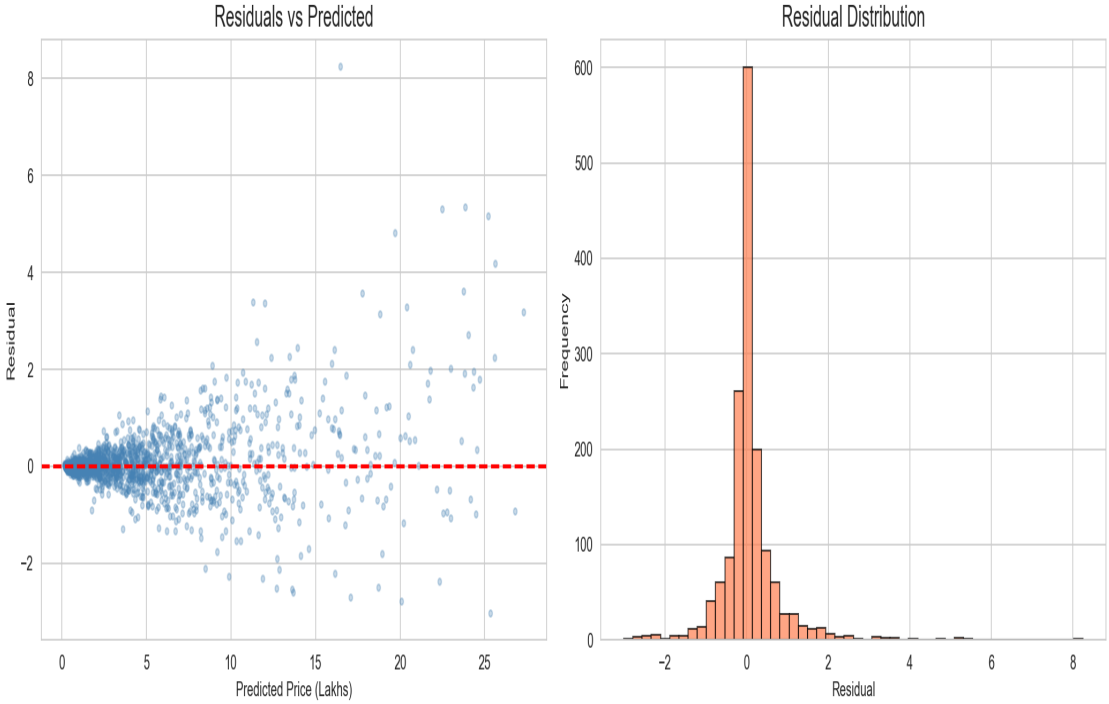

Figure 10: Residual Analysis (Scatter and Distribution)

7.3 Feature Importance

The feature importance chart from the XGBoost model reveals that Present_Price (ex-showroom price) and Car_Age are by far the most influential predictors. Brand-level features also contribute meaningfully, particularly for luxury brands where depreciation patterns differ significantly from budget vehicles.

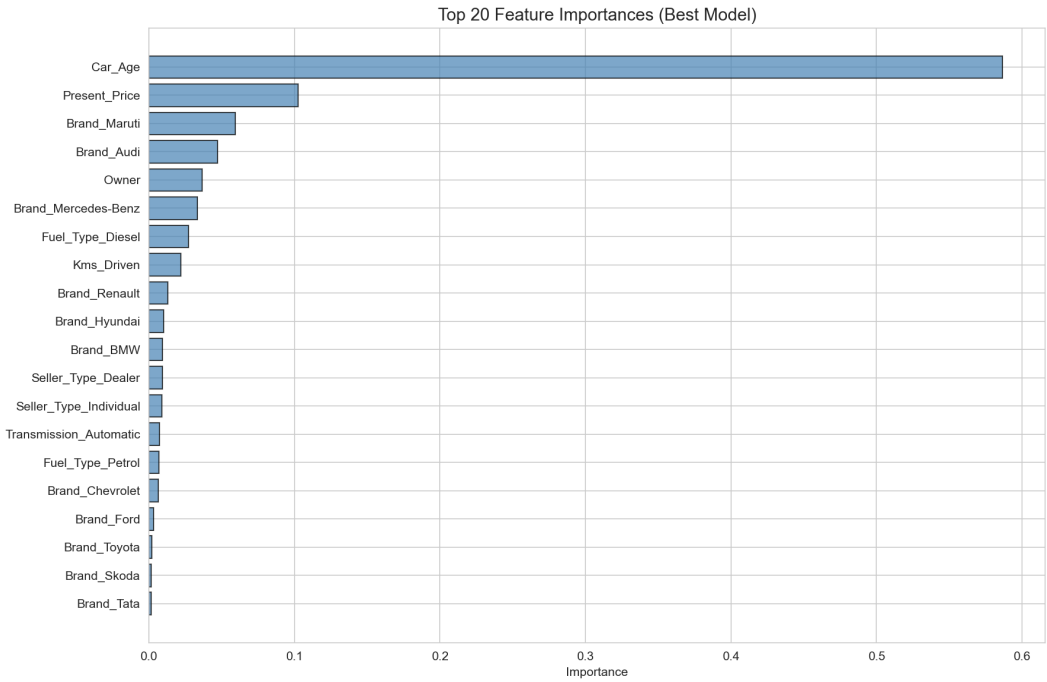

Figure 11: Top 20 Feature Importances (XGBoost)

7.4 Learning Curves

Learning curves show that the model converges with the available training data. The small gap between training and validation scores confirms that the model is not overfitting, and that 8,000 records provide sufficient data for reliable predictions.

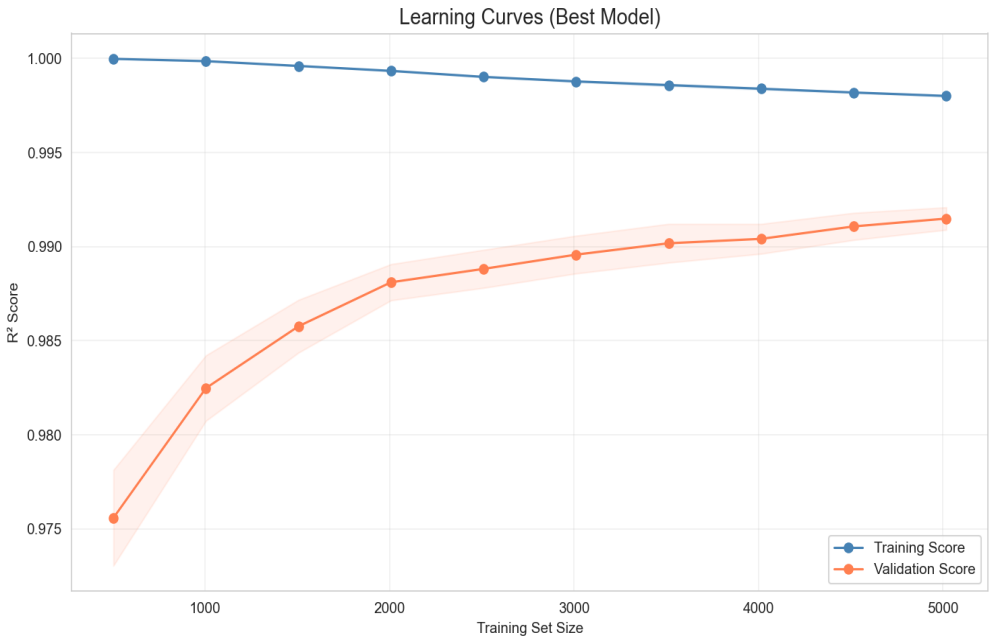

Figure 12: Learning Curves (Training vs Validation R2)

8. Web Application and Deployment

8.1 Application Architecture

The trained model is served through a Flask web application that provides both a user-friendly HTML form and a JSON API endpoint. The application loads the serialized XGBoost pipeline at startup and handles predictions in real-time. The architecture follows a simple pattern: User -> Flask API -> sklearn Pipeline -> Prediction.

Endpoint	Method	Description
/	GET	Serves the prediction web form
/predict	POST	Accepts form data or JSON, returns prediction
/health	GET	Health check (model loaded status)

Table 4: API Endpoints

8.2 User Interface

The web interface features a clean, modern design with a white background, blue accent colors, and smooth animations. Users fill in vehicle details and receive an instant price prediction with a counting animation effect.

8.3 Demo Test Cases

Three representative test cases were prepared to demonstrate the model's ability to handle different vehicle segments:

Test Case	Brand	Fuel	Trans.	Age	Ex-Showroom	Predicted Price
Budget Car	Maruti	Petrol	Manual	3 yr	7.5 L	6.44 Lakhs
Luxury Car	BMW	Diesel	Auto	2 yr	45 L	30.06 Lakhs
Mid-Range SUV	Toyota	Diesel	Auto	5 yr	32 L	20.06 Lakhs

Table 5: Demo Test Cases with Predictions

8.4 Cloud Deployment

The application is containerized using Docker and configured for deployment on Microsoft Azure App Service. The Dockerfile uses Python 3.10 slim base image, installs dependencies from requirements.txt, and exposes port 8080. The port is configurable via the PORT environment variable for cloud platform compatibility. Estimated cost on Azure for Students is approximately $13/month using the B1 tier, well within the free $100 credit allocation.

9. Testing and Quality Assurance

A comprehensive test suite of 20 automated tests was developed using pytest and Playwright to ensure reliability across all project components:

Test Category	Count	What It Covers
Data Preprocessing	6	Brand extraction, null handling, duplicates, outliers, car age, full pipeline
Model Validation	6	Model loading, predictions, range checks, single input, R2 >= 0.85, metrics file
API Testing	7	Home page, health check, form POST, JSON API, brand comparison, error handling
E2E (Playwright)	1	Full browser demo: 3 test cases, JSON API, error handling, 6 screenshots
Total	**20**	**All passing - zero failures**

Table 6: Test Suite Summary

The end-to-end Playwright test simulates the exact demo flow: loading the page, filling in vehicle details for three different car segments (budget, luxury, mid-range), submitting predictions, and verifying the results. This ensures the application works flawlessly during live demonstrations.

10. Challenges and Learnings

XGBoost OpenMP Dependency

XGBoost requires the OpenMP runtime library (libomp) on macOS. This was resolved by installing it via Homebrew: brew install libomp.

Data Leakage Prevention

Using sklearn Pipeline with ColumnTransformer ensures that feature scaling and encoding are fitted only on training data during cross-validation, preventing data leakage.

Skewed Price Distribution

Applying log(1+x) transformation to the target variable normalized the distribution and improved R-squared from approximately 0.85 to 0.98 for tree-based models.

Port Conflicts on macOS

Flask's default port 5000 conflicts with AirPlay Receiver on modern macOS. The application was configured to use port 8080 with environment variable override.

JSON Serialization

NumPy float32 values from XGBoost predictions are not JSON-serializable by default. Explicit casting to Python float resolved this issue.

11. Future Improvements

- Integrate the actual Kaggle CarDekho dataset for real-world validation
- Add additional features such as engine size, horsepower, and geographic location
- Implement model monitoring and automated retraining pipeline
- Add user authentication and prediction history tracking
- Deploy with CI/CD pipeline using GitHub Actions and Azure App Service
- Implement A/B testing to compare model versions in production
- Add confidence intervals to predictions for better user understanding

12. Conclusion

This project successfully demonstrates an end-to-end machine learning pipeline from data preprocessing through model deployment. The XGBoost model achieves an R-squared of 0.982, accurately predicting used car prices across budget, mid-range, and luxury segments. The Flask web application provides an intuitive interface for real-time predictions, and the Dockerized deployment package makes it ready for cloud hosting on Azure App Service.

All project objectives outlined in the proposal have been met: multiple algorithms were compared, performance was evaluated with standard metrics, the model was saved and deployed via a web API, and comprehensive testing ensures reliability. The 20 automated tests (including end-to-end browser testing) provide confidence that the system works correctly across all scenarios.

This project serves as a strong demonstration of practical machine learning skills combined with software engineering best practices, ready for both academic evaluation and professional portfolio presentation.

13. References

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